Assignment No: 5

Databases-SQLite, Messaging and E-mail

1. Create table Company (id, name, address, phno). Create Application for Performing the following operation on the table.
a) Insert New Company Details.
b) Show All the Company Details.

Where Is the Table Created?

The table is created inside the:

`DatabaseHelper.java` → `onCreate()` method

```
@Override
public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE Company (id INTEGER PRIMARY KEY AUTOINCREMENT,
name TEXT, address TEXT, phno TEXT)");
}
```

When Does This Execute?

The table is created automatically when:

```
myDb = new DatabaseHelper(this);
```

is executed for the **first time** in `MainActivity`.

Android then:

1. Creates the database file (`CompanyDB.db`)
2. Calls `onCreate()`
3. Executes the `CREATE TABLE` statement

Where Is the Database Stored?

Android stores it internally at:

```
/data/data/your.package.name/databases/CompanyDB.db
```

You don't need to manually create it.

Important

You **do not create the table separately in Android Studio**.
It is automatically created when the app runs the first time.

If Table Is Not Updating?

If you modify the table structure:

- Uninstall the app from emulator/device
  OR
- Increase database version in constructor:

```
super(context, DATABASE_NAME, null, 2);
```

This triggers `onUpgrade()`.

SQL: Create Company Table

```
CREATE TABLE Company (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    address TEXT NOT NULL,
    phno TEXT NOT NULL
);
```

In Android Application

We will create:

- `DatabaseHelper.java`
- `MainActivity.java`
- `activity_main.xml`

Create Database Helper Class

DatabaseHelper.java

```
package com.example.companyapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "CompanyDB.db";
    private static final String TABLE_NAME = "Company";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE Company (id INTEGER PRIMARY KEY
AUTOINCREMENT, name TEXT, address TEXT, phno TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Company");
        onCreate(db);
    }
```

```
    // Insert Data
    public boolean insertData(String name, String address, String phno) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put("name", name);
        contentValues.put("address", address);
        contentValues.put("phno", phno);

        long result = db.insert(TABLE_NAME, null, contentValues);
        return result != -1;
    }

    // Get All Data
    public Cursor getAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.rawQuery("SELECT * FROM Company", null);
    }
}
```

Design Layout

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:orientation="vertical"
        android:padding="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <EditText
            android:id="@+id/editName"
            android:hint="Company Name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editAddress"
            android:hint="Address"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/editPhone"
            android:hint="Phone Number"
            android:inputType="phone"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnInsert"
            android:text="Insert Company"
            android:layout_width="match_parent"
```

```
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnView"
            android:text="View All Companies"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.companyapp;

import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper myDb;
    EditText editName, editAddress, editPhone;
    Button btnInsert, btnView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        myDb = new DatabaseHelper(this);

        editName = findViewById(R.id.editName);
        editAddress = findViewById(R.id.editAddress);
        editPhone = findViewById(R.id.editPhone);
        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);

        insertData();
        viewAll();
    }

    public void insertData() {
        btnInsert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                boolean isInserted = myDb.insertData(
                        editName.getText().toString(),
                        editAddress.getText().toString(),
                        editPhone.getText().toString()
                );

                if(isInserted)
```

```
                        Toast.makeText(MainActivity.this, "Data Inserted",
Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(MainActivity.this, "Data Not Inserted",
Toast.LENGTH_LONG).show();
            }
        });
    }

    public void viewAll() {
        btnView.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                Cursor res = myDb.getAllData();
                if(res.getCount() == 0) {
                    showMessage("Error", "No Data Found");
                    return;
                }

                StringBuffer buffer = new StringBuffer();
                while(res.moveToNext()) {
                    buffer.append("ID: " + res.getString(0) + "\n");
                    buffer.append("Name: " + res.getString(1) + "\n");
                    buffer.append("Address: " + res.getString(2) + "\n");
                    buffer.append("Phone: " + res.getString(3) + "\n\n");
                }

                showMessage("Company Details", buffer.toString());
            }
        });
    }

    public void showMessage(String title, String message) {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setCancelable(true);
        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
}
```

2. Create Table Emp (emp_no,emp_name,address,phone,salary), Dept(dept_no,dept_name,location), Emp-Dept is related with one-many relationship. Create an application for performing the following Operation on the table
a) Add Records into Emp and Dept table.
b) Accept Department name from User and delete employee information which belongs to that department.

Project Components

- `MainActivity.java`
- `DBHelper.java` (SQLiteOpenHelper class)
- `activity_main.xml`

DBHelper Class (Database Operations)

DBHelper.java

```
package com.example.empdeptapp;
import android.content.ContentValues;
import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    private static final String DB_NAME = "CompanyDB";
    private static final int DB_VERSION = 1;

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE Dept (" +
                "dept_no INTEGER PRIMARY KEY," +
                "dept_name TEXT UNIQUE," +
                "location TEXT)");

        db.execSQL("CREATE TABLE Emp (" +
                "emp_no INTEGER PRIMARY KEY," +
                "emp_name TEXT," +
                "address TEXT," +
                "phone TEXT," +
                "salary REAL," +
                "dept_no INTEGER," +
                "FOREIGN KEY(dept_no) REFERENCES Dept(dept_no))");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Emp");
        db.execSQL("DROP TABLE IF EXISTS Dept");
```

```
        onCreate(db);
    }

    // Insert Department
    public boolean insertDept(int deptNo, String deptName, String location)
{
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("dept_no", deptNo);
        cv.put("dept_name", deptName);
        cv.put("location", location);
        long result = db.insert("Dept", null, cv);
        return result != -1;
    }

    // Insert Employee
    public boolean insertEmp(int empNo, String empName, String address,
                             String phone, double salary, int deptNo) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("emp_no", empNo);
        cv.put("emp_name", empName);
        cv.put("address", address);
        cv.put("phone", phone);
        cv.put("salary", salary);
        cv.put("dept_no", deptNo);
        long result = db.insert("Emp", null, cv);
        return result != -1;
    }

    // Delete Employees by Department Name
    public int deleteEmpByDeptName(String deptName) {
        SQLiteDatabase db = this.getWritableDatabase();

        String query = "DELETE FROM Emp WHERE dept_no IN " +
                "(SELECT dept_no FROM Dept WHERE dept_name = ?)";

        return db.delete("Emp",
                "dept_no IN (SELECT dept_no FROM Dept WHERE dept_name = ?)",
                new String[]{deptName});
    }
}
```

---

activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etDeptNo"
        android:layout_height="50dp"
        android:layout_width="match_parent"
        android:hint="Department No"/>
```

```
<EditText
    android:id="@+id/etDeptName"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Department Name"/>

<EditText
    android:id="@+id/etLocation"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Location"/>

<Button
    android:id="@+id/btnAddDept"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:text="Add Department"/>

<!-- Employee Inputs -->
<EditText
    android:id="@+id/etEmpNo"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Employee No"/>

<EditText
    android:id="@+id/etEmpName"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Employee Name"/>

<EditText
    android:id="@+id/etAddress"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Address"/>

<EditText
    android:id="@+id/etPhone"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Phone"/>

<EditText
    android:id="@+id/etSalary"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Salary"/>

<EditText
    android:id="@+id/etEmpDeptNo"
    android:layout_height="50dp"
    android:layout_width="match_parent"
    android:hint="Department No"/>
```

```
    <Button
        android:id="@+id/btnAddEmp"
        android:layout_height="50dp"
        android:layout_width="match_parent"
        android:text="Add Employee"/>

    <!-- Delete Section -->
    <EditText
        android:id="@+id/etDeleteDept"
        android:layout_height="50dp"
        android:layout_width="match_parent"
        android:hint="Enter Dept Name to Delete Employees"/>

    <Button
        android:id="@+id/btnDelete"
        android:layout_height="50dp"
        android:layout_width="match_parent"
        android:text="Delete Employees"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.empdeptapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    DBHelper dbHelper;

    EditText etDeptNo, etDeptName, etLocation;
    EditText etEmpNo, etEmpName, etAddress, etPhone, etSalary, etEmpDeptNo;
    EditText etDeleteDept;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        dbHelper = new DBHelper(this);

        etDeptNo = findViewById(R.id.etDeptNo);
        etDeptName = findViewById(R.id.etDeptName);
        etLocation = findViewById(R.id.etLocation);
        etEmpNo = findViewById(R.id.etEmpNo);
        etEmpName = findViewById(R.id.etEmpName);
        etAddress = findViewById(R.id.etAddress);
        etPhone = findViewById(R.id.etPhone);
        etSalary = findViewById(R.id.etSalary);
        etEmpDeptNo = findViewById(R.id.etEmpDeptNo);

        etDeleteDept = findViewById(R.id.etDeleteDept);

        Button btnAddDept = findViewById(R.id.btnAddDept);
        Button btnAddEmp = findViewById(R.id.btnAddEmp);
        Button btnDelete = findViewById(R.id.btnDelete);
```

```
        // Add Department
        btnAddDept.setOnClickListener(v -> {
            boolean result = dbHelper.insertDept(
                    Integer.parseInt(etDeptNo.getText().toString()),
                    etDeptName.getText().toString(),
                    etLocation.getText().toString());

            Toast.makeText(this,
                    result ? "Department Added" : "Failed",
                    Toast.LENGTH_SHORT).show();
        });

        // Add Employee
        btnAddEmp.setOnClickListener(v -> {
            boolean result = dbHelper.insertEmp(
                    Integer.parseInt(etEmpNo.getText().toString()),
                    etEmpName.getText().toString(),
                    etAddress.getText().toString(),
                    etPhone.getText().toString(),
                    Double.parseDouble(etSalary.getText().toString()),
                    Integer.parseInt(etEmpDeptNo.getText().toString()));

            Toast.makeText(this,
                    result ? "Employee Added" : "Failed",
                    Toast.LENGTH_SHORT).show();
        });

        // Delete Employees by Department Name
        btnDelete.setOnClickListener(v -> {
            int count = dbHelper.deleteEmpByDeptName(
                    etDeleteDept.getText().toString());

               Toast.makeText(this, count + " Employees Deleted",
Toast.LENGTH_SHORT).show();
        });
    }
}
```

3) Create application to send and receive messages using SMSManager.

1 Create New Project

- Open Android Studio
- Select **Empty Activity**
- Language: **Java**
- Minimum SDK: API 21+

---

2 Add Permissions (AndroidManifest.xml)

```
<uses-permission android:name="android.permission.SEND_SMS"/>
<uses-permission android:name="android.permission.RECEIVE_SMS"/>
<uses-permission android:name="android.permission.READ_SMS"/>

<application
    ... >

    <receiver android:name=".SmsReceiver">
        <intent-filter>
            <action
android:name="android.provider.Telephony.SMS_RECEIVED"/>
        </intent-filter>
    </receiver>

</application>
```

---

3 Design Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etPhone"
        android:hint="Enter Phone Number"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etMessage"
        android:hint="Enter Message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSend"
        android:text="Send SMS"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
```

```
        android:id="@+id/tvReceived"
        android:text="Received Messages"
        android:paddingTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

---

4. MainActivity.java (Send SMS)

```
package com.example.smsapp;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

public class MainActivity extends AppCompatActivity {

    EditText etPhone, etMessage;
    Button btnSend;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etPhone = findViewById(R.id.etPhone);
        etMessage = findViewById(R.id.etMessage);
        btnSend = findViewById(R.id.btnSend);

        // Request Runtime Permission
        ActivityCompat.requestPermissions(this,
                new String[]{Manifest.permission.SEND_SMS,
                        Manifest.permission.RECEIVE_SMS,
                        Manifest.permission.READ_SMS}, 1);

        btnSend.setOnClickListener(v -> {
            String phone = etPhone.getText().toString();
            String message = etMessage.getText().toString();

            try {
                SmsManager smsManager = SmsManager.getDefault();
                smsManager.sendTextMessage(phone, null, message, null,
null);
                Toast.makeText(this, "SMS Sent Successfully",
Toast.LENGTH_SHORT).show();
            } catch (Exception e) {
                Toast.makeText(this, "SMS Failed to Send",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

5 Create SmsReceiver.java (Receive SMS)

```
package com.example.smsapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.telephony.SmsMessage;
import android.widget.Toast;

public class SmsReceiver extends BroadcastReceiver {

    @Override
    public void onReceive(Context context, Intent intent) {

        Bundle bundle = intent.getExtras();
        if (bundle != null) {

            Object[] pdus = (Object[]) bundle.get("pdus");

            for (Object pdu : pdus) {

                SmsMessage message = SmsMessage.createFromPdu((byte[])
pdu);
                String sender = message.getOriginatingAddress();
                String body = message.getMessageBody();

                Toast.makeText(context,
                        "SMS from: " + sender + "\n" + body,
                        Toast.LENGTH_LONG).show();
            }
        }
    }
}
```

---

6 How It Works (Concept)

Sending SMS Flow:

```
User → App → SmsManager → SIM → Mobile Network → Receiver
```

Receiving SMS Flow:

```
Mobile Network → Android System → Broadcast → SmsReceiver
```

Q3: Create application to send email.

1 Create New Project

- Open Android Studio
- Select **Empty Activity**
- Language: **Java**
- Minimum SDK: API 21+

---

2 Design Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etEmail"
        android:hint="Recipient Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etSubject"
        android:hint="Subject"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etMessage"
        android:hint="Message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSend"
        android:text="Send Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

---

3 MainActivity.java

```
package com.example.emailapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
```

```
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etEmail, etSubject, etMessage;
    Button btnSend;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etEmail = findViewById(R.id.etEmail);
        etSubject = findViewById(R.id.etSubject);
        etMessage = findViewById(R.id.etMessage);
        btnSend = findViewById(R.id.btnSend);

        btnSend.setOnClickListener(v -> {

            String email = etEmail.getText().toString();
            String subject = etSubject.getText().toString();
            String message = etMessage.getText().toString();

            Intent intent = new Intent(Intent.ACTION_SEND);
            intent.setType("message/rfc822");
            intent.putExtra(Intent.EXTRA_EMAIL, new String[]{email});
            intent.putExtra(Intent.EXTRA_SUBJECT, subject);
            intent.putExtra(Intent.EXTRA_TEXT, message);

            try {
                startActivity(Intent.createChooser(intent, "Choose Email
Client"));
            } catch (Exception e) {
                Toast.makeText(this, "No Email App Installed",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

How It Works

- Intent.ACTION_SEND
  Tells Android we want to send data.
- setType("message/rfc822")
  Filters only email apps.
- EXTRA_EMAIL
  Recipient address.
- EXTRA_SUBJECT
  Email subject.
- EXTRA_TEXT
  Email body.

Email Sending Flow

User → Your App → Intent → Email App (Gmail etc.) → Recipient

Set B) 1. Create sample application with login module (Check username and password). On successful login, pass username to next screen And on failing login, alert user using Toast (Hint: Use Login(username, password) Table.)

Project Structure

```
MainActivity.java      → Login Screen
HomeActivity.java      → Next Screen (after login)
DBHelper.java          → SQLite Database Helper
activity_main.xml      → Login UI
activity_home.xml      → Home UI
```

1 Create New Project

- Android Studio → Empty Activity
- Language: Java

2 Create Database Helper Class

DBHelper.java

```
package com.example.loginapp;

import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public DBHelper(Context context) {
        super(context, "LoginDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE Login(username TEXT PRIMARY KEY, password
TEXT)");

        // Insert sample user
        db.execSQL("INSERT INTO Login VALUES('admin','1234')");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Login");
        onCreate(db);
    }

    // Check Login Method
    public boolean checkLogin(String username, String password) {
```

```
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery(
                "SELECT * FROM Login WHERE username=? AND password=?",
                new String[]{username, password});

        if (cursor.getCount() > 0) {
            cursor.close();
            return true;
        } else {
            cursor.close();
            return false;
        }
    }
}
```

3 Login Screen Layout

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etUsername"
        android:hint="Username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

4 MainActivity.java (Login Logic)

```
package com.example.loginapp;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
```

```
    Button btnLogin;
    DBHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        dbHelper = new DBHelper(this);

        btnLogin.setOnClickListener(v -> {

            String username = etUsername.getText().toString();
            String password = etPassword.getText().toString();

            boolean isValid = dbHelper.checkLogin(username, password);

            if (isValid) {
                Intent intent = new Intent(MainActivity.this,
HomeActivity.class);
                intent.putExtra("username", username);
                startActivity(intent);
            } else {
                Toast.makeText(MainActivity.this,
                        "Invalid Username or Password",
                        Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

5 Home Screen Layout

activity_home.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/tvWelcome"
        android:textSize="20sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

6 HomeActivity.java

```
package com.example.loginapp;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class HomeActivity extends AppCompatActivity {
```

```
    TextView tvWelcome;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_home);

        tvWelcome = findViewById(R.id.tvWelcome);

        String username = getIntent().getStringExtra("username");

        tvWelcome.setText("Welcome " + username);
    }
}
```

Set B) 2 Create Tables:

Project(pno, p_name, ptype, duration)
Employee(id, e_name, qualification, joindate)
_Employee (Many-to-Many Relationship Table)

- Add new records
- Accept project name from user
- Display employees working on that project

1 Database Design

Since **Project ↔ Employee** is **Many-to-Many**, we must create a **junction table**.

Tables:

Project

Column	Type
Pno	INTEGER PRIMARY KEY
p_name	TEXT
Ptype	TEXT
Duration	TEXT

Employee

Column	Type
Id	INTEGER PRIMARY KEY
e_name	TEXT
qualification	TEXT
Joindate	TEXT

Project_Employee (Junction Table)

Column	Type
Pno	INTEGER
Id	INTEGER

This table links both tables.

2 Create DBHelper.java

```
package com.example.projectapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public DBHelper(Context context) {
        super(context, "CompanyDB", null, 1);
    }

    @Override
```

```
    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE Project(pno INTEGER PRIMARY KEY, p_name
TEXT, ptype TEXT, duration TEXT)");

        db.execSQL("CREATE TABLE Employee(id INTEGER PRIMARY KEY, e_name
TEXT, qualification TEXT, joindate TEXT)");

        db.execSQL("CREATE TABLE Project_Employee(pno INTEGER, id
INTEGER)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {
        db.execSQL("DROP TABLE IF EXISTS Project");
        db.execSQL("DROP TABLE IF EXISTS Employee");
        db.execSQL("DROP TABLE IF EXISTS Project_Employee");
        onCreate(db);
    }

    // Insert Project
    public boolean insertProject(int pno, String name, String type, String
duration) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("pno", pno);
        cv.put("p_name", name);
        cv.put("ptype", type);
        cv.put("duration", duration);
        long result = db.insert("Project", null, cv);
        return result != -1;
    }

    // Insert Employee
    public boolean insertEmployee(int id, String name, String
qualification, String joindate) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("id", id);
        cv.put("e_name", name);
        cv.put("qualification", qualification);
        cv.put("joindate", joindate);
        long result = db.insert("Employee", null, cv);
        return result != -1;
    }

    // Assign Employee to Project
    public void assignEmployee(int pno, int id) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("pno", pno);
        cv.put("id", id);
        db.insert("Project_Employee", null, cv);
    }

    // Get Employees by Project Name
    public Cursor getEmployeesByProject(String projectName) {
        SQLiteDatabase db = this.getReadableDatabase();
```

```
        String query = "SELECT Employee.e_name, Employee.qualification,
Employee.joindate " +
                "FROM Employee " +
                "INNER JOIN Project_Employee ON Employee.id =
Project_Employee.id " +
                "INNER JOIN Project ON Project.pno = Project_Employee.pno "
+
                "WHERE Project.p_name = ?";

        return db.rawQuery(query, new String[]{projectName});
    }
}
```

3 Layout (activity_main.xml)

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etProjectName"
        android:hint="Enter Project Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSearch"
        android:text="Search Employees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="20dp"/>

</LinearLayout>
```

4 MainActivity.java

```
package com.example.projectapp;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    DBHelper dbHelper;
    EditText etProjectName;
    Button btnSearch;
    TextView tvResult;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        dbHelper = new DBHelper(this);

        etProjectName = findViewById(R.id.etProjectName);
        btnSearch = findViewById(R.id.btnSearch);
        tvResult = findViewById(R.id.tvResult);

        btnSearch.setOnClickListener(v -> {

            String projectName = etProjectName.getText().toString();
            Cursor cursor = dbHelper.getEmployeesByProject(projectName);

            if (cursor.getCount() == 0) {
                tvResult.setText("No employees found.");
            } else {
                StringBuilder builder = new StringBuilder();
                while (cursor.moveToNext()) {
                    builder.append("Name: ")
                            .append(cursor.getString(0))
                            .append("\nQualification: ")
                            .append(cursor.getString(1))
                            .append("\nJoin Date: ")
                            .append(cursor.getString(2))
                            .append("\n\n");
                }
                tvResult.setText(builder.toString());
            }
            cursor.close();
        });
    }
}
```

Set B. 3) Create an application to design login form, validate it. Write and send email with appropriate message.

Design Login Form (XML Layout)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="16dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="24dp"/>

</LinearLayout>
```

Add Validation and Email Code (MainActivity.java)

```
package com.example.loginapp;

import android.content.Intent;
import android.os.Bundle;
import android.util.Patterns;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etEmail, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etEmail = findViewById(R.id.etEmail);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                validateAndSendEmail();
            }
        });
    }

    private void validateAndSendEmail() {

        String email = etEmail.getText().toString().trim();
        String password = etPassword.getText().toString().trim();

        // Validation
        if (email.isEmpty()) {
            etEmail.setError("Email is required");
            etEmail.requestFocus();
            return;
        }

        if (!Patterns.EMAIL_ADDRESS.matcher(email).matches()) {
            etEmail.setError("Enter valid email");
            etEmail.requestFocus();
            return;
        }

        if (password.isEmpty()) {
            etPassword.setError("Password is required");
            etPassword.requestFocus();
            return;
        }

        if (password.length() < 6) {
            etPassword.setError("Password must be at least 6 characters");
            etPassword.requestFocus();
            return;
        }

        Toast.makeText(this, "Login Successful!",
Toast.LENGTH_SHORT).show();

        sendEmail(email);
    }

    private void sendEmail(String userEmail) {

        Intent intent = new Intent(Intent.ACTION_SEND);
        intent.setType("message/rfc822");
        intent.putExtra(Intent.EXTRA_EMAIL, new
String[]{"admin@example.com"});
        intent.putExtra(Intent.EXTRA_SUBJECT, "Login Successful");
        intent.putExtra(Intent.EXTRA_TEXT,
                "User with email " + userEmail + " has logged in
successfully.");
```

```
        try {
            startActivity(Intent.createChooser(intent, "Send Email"));
        } catch (Exception e) {
            Toast.makeText(this, "No email app found!",
Toast.LENGTH_SHORT).show();
        }
    }
}
```

✓ Validation Checks

- Email not empty
- Valid email format
- Password not empty
- Password minimum 6 characters

✓ Email Sending

- Opens installed email app (like Gmail)
- Sends message:
    - Subject: *Login Successful*
    - Message: *User with email ___ has logged in successfully*

Output Flow

1. User enters email + password
2. Clicks Login
3. If valid → Shows success message
4. Opens email app with prefilled message

5. **Create an application to send SMS with image and contact as attachment**

To build a simple Android app that sends an **SMS with an image and a contact attached**, you need to use **MMS (Multimedia Messaging Service)** — because normal SMS does not support attachments.

Important Notes

- SMS = Text only
- MMS = Text + Image + Contact
- Modern Android (Android 10+) does **not allow direct MMS sending** without user interaction.
- So we will open the default SMS app with attachments pre-filled.

Add Permissions (AndroidManifest.xml)

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.smsapp">

    <uses-permission android:name="android.permission.READ_CONTACTS"/>
    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"/>

    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Layout Design (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"/>
```

```
    <Button
        android:id="@+id/btnSelectImage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Image"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnSelectContact"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Contact"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send MMS"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.smsapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.ContactsContract;
import android.provider.MediaStore;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etPhone;
    Button btnSelectImage, btnSelectContact, btnSend;

    Uri imageUri = null;
    Uri contactUri = null;

    static final int IMAGE_REQUEST = 1;
    static final int CONTACT_REQUEST = 2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etPhone = findViewById(R.id.etPhone);
        btnSelectImage = findViewById(R.id.btnSelectImage);
        btnSelectContact = findViewById(R.id.btnSelectContact);
        btnSend = findViewById(R.id.btnSend);
```

```
        btnSelectImage.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_PICK,
                    MediaStore.Images.Media.EXTERNAL_CONTENT_URI);
            startActivityForResult(intent, IMAGE_REQUEST);
        });

        btnSelectContact.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_PICK,
                    ContactsContract.Contacts.CONTENT_URI);
            startActivityForResult(intent, CONTACT_REQUEST);
        });

        btnSend.setOnClickListener(v -> sendMMS());
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent
data) {
        super.onActivityResult(requestCode, resultCode, data);

        if (resultCode == RESULT_OK && data != null) {
            if (requestCode == IMAGE_REQUEST) {
                imageUri = data.getData();
                Toast.makeText(this, "Image Selected",
Toast.LENGTH_SHORT).show();
            }

            if (requestCode == CONTACT_REQUEST) {
                contactUri = data.getData();
                Toast.makeText(this, "Contact Selected",
Toast.LENGTH_SHORT).show();
            }
        }
    }

    private void sendMMS() {

        String phone = etPhone.getText().toString().trim();

        if (phone.isEmpty()) {
            etPhone.setError("Enter phone number");
            return;
        }

        if (imageUri == null) {
            Toast.makeText(this, "Select image first",
Toast.LENGTH_SHORT).show();
            return;
        }

        Intent intent = new Intent(Intent.ACTION_SEND_MULTIPLE);
        intent.setType("image/*");
        intent.putExtra("address", phone);
        intent.putExtra("sms_body", "Please see attached image and
contact.");

        intent.putParcelableArrayListExtra(Intent.EXTRA_STREAM,
                new java.util.ArrayList<Uri>() {{
                    add(imageUri);
                    if (contactUri != null) add(contactUri);
                }});
```

```
        intent.setPackage("com.android.mms");

        try {
            startActivity(intent);
        } catch (Exception e) {
            Toast.makeText(this, "No messaging app found",
Toast.LENGTH_SHORT).show();
        }
    }
}
```

Working:

1. Enter phone number
2. Select image
3. Select contact
4. Click Send MMS
5. Default messaging app opens with:
    - Phone number
    - Text message
    - Image attached
    - Contact attached (if selected)

Important (Android 11+)

- `startActivityForResult()` is deprecated
- For production apps, use `ActivityResultLauncher`
- Some devices may block direct MMS sending
- App must be default SMS app for full control

Output

✓ Sends MMS
✓ Image attached
✓ Contact attached
✓ Works on most devices

Assignment VI: Location-Based Services and Google Map

Set A:

1) Write a program to find the specific location of an Android device and display details of the place like Address line, city with Geocoding.

Add Permissions (AndroidManifest.xml)

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.locationapp">

    <uses-permission
android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <uses-permission
android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <uses-permission android:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Design Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnLocation"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Get Current Location"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Location details will appear here"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.locationapp;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Address;
import android.location.Geocoder;
import android.location.Location;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationServices;

import java.io.IOException;
import java.util.List;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    private FusedLocationProviderClient fusedLocationClient;
    TextView tvResult;
    Button btnLocation;

    private static final int LOCATION_REQUEST = 100;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);
        btnLocation = findViewById(R.id.btnLocation);

        fusedLocationClient =
LocationServices.getFusedLocationProviderClient(this);

        btnLocation.setOnClickListener(v -> getLocation());
    }

    private void getLocation() {

        if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.ACCESS_FINE_LOCATION)
                != PackageManager.PERMISSION_GRANTED) {

    ActivityCompat.requestPermissions(this,
           new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
                    LOCATION_REQUEST);
            return;
        }

 fusedLocationClient.getLastLocation()
```

```
                .addOnSuccessListener(this, location -> {
                    if (location != null) {
                        getAddress(location);
                    } else {
                     Toast.makeText(this, "Unable to get location",
                                Toast.LENGTH_SHORT).show();
                    }
                });
    }

    private void getAddress(Location location) {

        Geocoder geocoder = new Geocoder(this, Locale.getDefault());

        try {
        List<Address> addresses = geocoder.getFromLocation(
                    location.getLatitude(),
                    location.getLongitude(),1);

            if (addresses != null && !addresses.isEmpty()) {

                Address address = addresses.get(0);
String details =
"Latitude: " + location.getLatitude() + "\n" +
"Longitude: " + location.getLongitude() + "\n\n" +
"Address Line: " + address.getAddressLine(0) + "\n" +
"City: " + address.getLocality() + "\n" +
"State: " + address.getAdminArea() + "\n" +
"Country: " + address.getCountryName() + "\n" +
"Postal Code: " + address.getPostalCode();

                tvResult.setText(details);
            }
        } catch (IOException e) {
            e.printStackTrace();
         Toast.makeText(this, "Geocoder failed",Toast.LENGTH_SHORT).show();
        }
    }

    @Override
public void onRequestPermissionsResult(int requestCode,@NonNull String[]
permissions, @NonNull int[] grantResults)
{
        super.onRequestPermissionsResult(requestCode, permissions,
grantResults);

        if (requestCode == LOCATION_REQUEST
                && grantResults.length > 0
                && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
            getLocation();
        }
    }
}
```

Working

1. User clicks **Get Current Location**
2. App checks permission
3. Gets Latitude & Longitude

4. Uses **Geocoder**
5. Displays:
    - Address Line
    - City
    - State
    - Country
    - Postal Code

Important Notes

- Location must be enabled on device
- Internet required for Geocoder (in many devices)
- On Android 11+, you must request runtime permission

---

2) Write a program to search a specific location on Google Map.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etLocation"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter location to search"
        android:inputType="text"/>

    <Button
        android:id="@+id/btnSearch"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Search on Map"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.mapsearchapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    EditText etLocation;
    Button btnSearch;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etLocation = findViewById(R.id.etLocation);
        btnSearch = findViewById(R.id.btnSearch);

        btnSearch.setOnClickListener(v -> openMap());
    }

    private void openMap() {

        String location = etLocation.getText().toString().trim();

        if (location.isEmpty()) {
            etLocation.setError("Enter location");
            return;
        }

        // Encode location for URL
        Uri uri = Uri.parse("geo:0,0?q=" + Uri.encode(location));

        Intent intent = new Intent(Intent.ACTION_VIEW, uri);
        intent.setPackage("com.google.android.apps.maps");

        try {
            startActivity(intent);
        } catch (Exception e) {
            Toast.makeText(this,
                    "Google Maps not installed",
                    Toast.LENGTH_SHORT).show();
        }
    }
}
```

Working:

1. User enters location (Example: *New York*, *Eiffel Tower*, *Delhi*)
2. Clicks **Search on Map**
3. App opens **Google Maps**
4. The searched location is displayed on map

3) Write a program to perform Zoom In, Zoom Out operation and display Satellite view, Terrain view of current location on Google Map.

Create New Google map Project

- Open Android Studio
- New Project → **Google Maps Activity**
- Language: Java

This automatically:

- Adds Google Maps dependency
- Creates `google_maps_api.xml`

Get Google Maps API Key

1. Go to **Google Cloud Console**
2. Enable **Maps SDK for Android**
3. Generate API key
4. Paste it inside:

```
res/values/google_maps_api.xml
<string name="google_maps_key">YOUR_API_KEY</string>
```

AndroidManifest.xml

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

Inside `<application>`:

```
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value="@string/google_maps_key"/>
```

Modify Layout (activity_maps.xml)

Add buttons below the map fragment:

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <fragment
        android:id="@+id/map"
        android:name="com.google.android.gms.maps.SupportMapFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <LinearLayout
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_alignParentBottom="true"
        android:background="#AAFFFFFF"
        android:padding="5dp">

        <Button
            android:id="@+id/btnZoomIn"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Zoom In"/>

        <Button
            android:id="@+id/btnZoomOut"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Zoom Out"/>

        <Button
            android:id="@+id/btnSatellite"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Satellite"/>

        <Button
            android:id="@+id/btnTerrain"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Terrain"/>

    </LinearLayout>

</RelativeLayout>
```

MapsActivity.java

```
package com.example.mapdemo;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.os.Bundle;
import android.widget.Button;

import androidx.annotation.NonNull;
import androidx.core.app.ActivityCompat;
import androidx.fragment.app.FragmentActivity;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationServices;
import com.google.android.gms.maps.*;
import com.google.android.gms.maps.model.*;

public class MapsActivity extends FragmentActivity implements
OnMapReadyCallback {

    private GoogleMap mMap;
    private FusedLocationProviderClient fusedLocationClient;
    private static final int LOCATION_REQUEST = 100;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_maps);

        fusedLocationClient =
LocationServices.getFusedLocationProviderClient(this);

        SupportMapFragment mapFragment =
                (SupportMapFragment) getSupportFragmentManager()
                        .findFragmentById(R.id.map);

        mapFragment.getMapAsync(this);
    }

    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        enableLocation();

        Button zoomIn = findViewById(R.id.btnZoomIn);
        Button zoomOut = findViewById(R.id.btnZoomOut);
        Button satellite = findViewById(R.id.btnSatellite);
        Button terrain = findViewById(R.id.btnTerrain);

        zoomIn.setOnClickListener(v ->
                mMap.animateCamera(CameraUpdateFactory.zoomIn()));

        zoomOut.setOnClickListener(v ->
                mMap.animateCamera(CameraUpdateFactory.zoomOut()));

        satellite.setOnClickListener(v ->
                mMap.setMapType(GoogleMap.MAP_TYPE_SATELLITE));

        terrain.setOnClickListener(v ->
                mMap.setMapType(GoogleMap.MAP_TYPE_TERRAIN));
    }

    private void enableLocation() {

        if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.ACCESS_FINE_LOCATION)
                != PackageManager.PERMISSION_GRANTED) {

            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
                    LOCATION_REQUEST);
            return;
        }

        mMap.setMyLocationEnabled(true);

        fusedLocationClient.getLastLocation()
                .addOnSuccessListener(this, location -> {
                    if (location != null) {
                        LatLng current =
                                new LatLng(location.getLatitude(),
                                           location.getLongitude());
```

```
                        mMap.moveCamera(
                                CameraUpdateFactory.newLatLngZoom(current,
15));
                    }
                });
    }

    @Override
    public void onRequestPermissionsResult(int requestCode,
                                           @NonNull String[] permissions,
                                           @NonNull int[] grantResults) {
        if (requestCode == LOCATION_REQUEST &&
                grantResults.length > 0 &&
                grantResults[0] == PackageManager.PERMISSION_GRANTED) {
            enableLocation();
        }
    }
}
```

Features Implemented

✓ Shows current location
✓ Zoom In button
✓ Zoom Out button
✓ Satellite view
✓ Terrain view

Set B.

1. Write a program to calculate distance between two locations on Google Map.

Create Project

- Open Android Studio
- New Project → **Google Maps Activity**
- Language: Java

Add Google Maps API Key

In:
`res/values/google_maps_api.xml`
Add your API key:
`<string name="google_maps_key">YOUR_API_KEY</string>`
Enable **Maps SDK for Android** in Google Cloud Console.

activity_maps.xml

Add a TextView to show distance.

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <fragment
        android:id="@+id/map"
        android:name="com.google.android.gms.maps.SupportMapFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

    <TextView
        android:id="@+id/tvDistance"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Distance: "
        android:textSize="18sp"
        android:background="#AAFFFFFF"
        android:padding="8dp"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"/>
</RelativeLayout>
```

MapsActivity.java

```
package com.example.distancemap;

import android.os.Bundle;
import android.widget.TextView;
```

```
import androidx.fragment.app.FragmentActivity;
import com.google.android.gms.maps.*;
import com.google.android.gms.maps.model.*;

public class MapsActivity extends FragmentActivity
        implements OnMapReadyCallback {

    private GoogleMap mMap;
    private LatLng point1 = null;
    private LatLng point2 = null;
    TextView tvDistance;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_maps);

        tvDistance = findViewById(R.id.tvDistance);

        SupportMapFragment mapFragment =
                (SupportMapFragment) getSupportFragmentManager()
                        .findFragmentById(R.id.map);

        mapFragment.getMapAsync(this);
    }

    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        mMap.setOnMapClickListener(latLng -> {

            if (point1 == null) {
                point1 = latLng;
                mMap.addMarker(new MarkerOptions()
                        .position(point1)
                        .title("Point 1"));

            } else if (point2 == null) {
                point2 = latLng;
                mMap.addMarker(new MarkerOptions()
                        .position(point2)
                        .title("Point 2"));

                drawLineAndCalculate();

            } else {
                // Reset points
                mMap.clear();
                point1 = latLng;
                point2 = null;
                tvDistance.setText("Distance: ");
                    mMap.addMarker(new
            MarkerOptions().position(point1).title("Point 1"));
            }
        });
    }

    private void drawLineAndCalculate() {
```

```
        mMap.addPolyline(new PolylineOptions()
                .add(point1, point2)
                .width(5)
                .color(android.graphics.Color.RED));

        float[] results = new float[1];
        android.location.Location.distanceBetween(
                point1.latitude, point1.longitude,
                point2.latitude, point2.longitude,
                results);

        float distanceInMeters = results[0];
        float distanceInKm = distanceInMeters / 1000;

        tvDistance.setText("Distance: " +
                String.format("%.2f", distanceInKm) + " km");
    }
}
```

Working:

1. User taps on map → First location selected
2. User taps again → Second location selected
3. App:
   - Adds two markers
   - Draws a red line between them
   - Calculates distance using `Location.distanceBetween()`
   - Displays distance in kilometers
4. Third tap → Resets map

Set B. 2)

To modify your program to **draw the path along the actual road route (not a straight line)**, you must use the **Directions API** instead of `Location.distanceBetween()`.

The straight red line you drew earlier is "as-the-crow-flies" distance.
To draw the real driving route, we will use:

Google Maps Directions API

What We Will Change

1. User selects two points
2. Call **Directions API**
3. Get route JSON response
4. Decode polyline
5. Draw real route on map

Enable Directions API

Go to **Google Cloud Console**

- Enable **Directions API**
- Use the same API key

Add Internet Permission

In `AndroidManifest.xml`

```
<uses-permission android:name="android.permission.INTERNET"/>
```

Replace drawLineAndCalculate() Method

We will:

- Make HTTP request
- Parse JSON
- Draw route polyline

Updated MapsActivity.java (Modified Code)

Add these imports:

```
import android.graphics.Color;
import android.os.AsyncTask;
import org.json.JSONArray;
import org.json.JSONObject;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
```

Replace drawLineAndCalculate() with:

```
private void drawRoute() {

    String url = getDirectionsUrl(point1, point2);
    new FetchURL().execute(url);
}
```

Create URL Method

```
private String getDirectionsUrl(LatLng origin, LatLng dest) {

    String str_origin = "origin=" + origin.latitude + "," +
origin.longitude;
    String str_dest = "destination=" + dest.latitude + "," +
dest.longitude;
    String mode = "mode=driving";
```

```
    String parameters = str_origin + "&" + str_dest + "&" + mode;
    String output = "json";

    String url = "https://maps.googleapis.com/maps/api/directions/"
            + output + "?" + parameters + "&key=YOUR_API_KEY";

    return url;
}
```

AsyncTask to Fetch Data

```
private class FetchURL extends AsyncTask<String, Void, String> {

    @Override
    protected String doInBackground(String... strings) {
        String data = "";
        try {
            URL url = new URL(strings[0]);
            HttpURLConnection connection =
                    (HttpURLConnection) url.openConnection();
            connection.connect();

            InputStream inputStream = connection.getInputStream();
            InputStreamReader reader =
                    new InputStreamReader(inputStream);

            int dataChar;
            while ((dataChar = reader.read()) != -1) {
                data += (char) dataChar;
            }

        } catch (Exception e) {
            e.printStackTrace();
        }
        return data;
    }

    @Override
    protected void onPostExecute(String s) {
        parseRoute(s);
    }
}
```

Parse JSON & Draw Route

```
private void parseRoute(String jsonData) {

    try {
        JSONObject jsonObject = new JSONObject(jsonData);
        JSONArray routes = jsonObject.getJSONArray("routes");

        JSONObject route = routes.getJSONObject(0);
        JSONObject overview_polyline =
                route.getJSONObject("overview_polyline");

        String encodedString =
                overview_polyline.getString("points");

        List<LatLng> list = decodePolyline(encodedString);

        PolylineOptions options = new PolylineOptions()
```

```
                .addAll(list)
                .width(10)
                .color(Color.BLUE);

        mMap.addPolyline(options);

    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

Decode Polyline Method

```
private List<LatLng> decodePolyline(String encoded) {

    List<LatLng> poly = new ArrayList<>();
    int index = 0, len = encoded.length();
    int lat = 0, lng = 0;

    while (index < len) {
        int b, shift = 0, result = 0;
        do {
            b = encoded.charAt(index++) - 63;
            result |= (b & 0x1f) << shift;
            shift += 5;
        } while (b >= 0x20);

        int dlat = ((result & 1) != 0 ?
                ~(result >> 1) : (result >> 1));
        lat += dlat;

        shift = 0;
        result = 0;
        do {
            b = encoded.charAt(index++) - 63;
            result |= (b & 0x1f) << shift;
            shift += 5;
        } while (b >= 0x20);

        int dlng = ((result & 1) != 0 ?
                ~(result >> 1) : (result >> 1));
        lng += dlng;

        LatLng p = new LatLng(
                lat / 1E5, lng / 1E5);
        poly.add(p);
    }

    return poly;
}
```

Important Change

Replace: `drawLineAndCalculate();`

With: `drawRoute();`